

WEEK 4 - ASSIGNMENT

Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

Assignment Tasks

Task 1: Created my first Azure Resource Group and understood how Azure resources can be organized under a common container.

- Explore Azure Portal
- Create a Resource Group :

Deliverable:

Screenshot of Resource Group

Create a resource group ...

Basics Tags Review + create

[Automation Link](#)

Basics

Subscription	Azure for Students
Resource group name	rg-week4-adf
Region	Central India

Tags

None

Final Deliverable:

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The main content area is titled 'Resource Manager | Resource groups'. On the left, there is a sidebar with navigation options: 'Resource Manager', 'All resources', 'Favorite resources', 'Recent resources', and 'Resource groups'. The 'Resource groups' section is expanded, showing a list of resource groups. The 'rg-week4-adf' resource group is selected and highlighted. The main pane displays the 'Overview' tab for the 'rg-week4-adf' resource group. It includes a search bar, a 'Create' button, and a 'Manage view' dropdown. Below this, there is a list of resources: 'Activity log', 'Access control (IAM)', 'Tags', 'Resource visualizer', 'Events', 'Settings', 'Cost Management', 'Monitoring', 'Automation', and 'Help'. The 'Resources' section shows a message: 'No resources match your filters. Try changing or clearing your filters.' The 'Create' button is highlighted in blue.

Description

I created a Resource Group named rg-week4-adf in the Central India region. This Resource Group will be used to organize all resources required for this assignment.

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Task 2: Storage Setup

Do:

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable:

- Screenshot of container with uploaded file

The screenshot shows the 'Allowed resource deployment regions' policy assignment in the Microsoft Azure portal. The page includes a header with navigation links and a search bar. Below the header, there are tabs for 'Parameters (1)', 'Resource selectors (0)', 'Overrides (0)', 'Exemptions (0)', 'Remediation (0)', 'Deployed resources', and 'Managed Identity'. The 'Parameters (1)' tab is active, showing a table with one parameter: 'listOfAllowedLocations' with the value '["malaysiawest","uaenorth","eastasia","southeastasia","kor..."]'. The table also shows the parameter name 'Allowed locations' and the policy assignment parameter reference type 'User defined parameter'.

Parameter ID	Parameter name	Parameter value	Policy assignment parameter reference type
listOfAllowedLocations	Allowed locations	["malaysiawest","uaenorth","eastasia","southeastasia","kor..."]	User defined parameter

Initially I selected Central India, but Azure returned a policy validation error. I investigated the Azure Policy assignments and found that my Azure for Students subscription only allows resource deployment in five predefined regions. After identifying the allowed locations, I changed the region accordingly and proceeded with deployment.

The screenshot shows the 'Create a storage account' wizard in the Microsoft Azure portal. The wizard is divided into several tabs: 'Basics', 'Advanced', 'Networking', 'Data protection', 'Security', 'Encryption', 'Tags', and 'Review + create'. The 'Basics' tab is active, showing the following configuration:

- Subscription: Azure for Students
- Resource group: rg-week4-adf
- Location: Southeast Asia
- Storage account name: stweek4yash2026
- Primary service: Azure Blob Storage or Azure Data Lake Storage
- Performance: Standard
- Replication: Locally redundant storage (LRS)

The 'Advanced' tab is also visible, showing options like 'Enable hierarchical namespace', 'Enable SFTP', 'Enable network file system v3', and 'Allow cross-tenant replication', all of which are disabled. At the bottom, there are buttons for 'Previous', 'Next', and 'Create', along with a 'Give feedback' link.

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Home

stweek4yash2026_1780314180664 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : stweek4yash2026_1780314180664

Subscription : Azure for Students

Resource group : rg-week4-adf

Start time : 6/1/2026, 5:14:20 PM

Correlation ID : ffb5832a-fb65-44f6-a3fc-34605bb1e7d8

Deployment details

Resource	Type	Status	Operation details
stweek4yash2026/	Microsoft.Storage/storageAccou	OK	Operation details
stweek4yash2026/	Microsoft.Storage/storageAccou	OK	Operation details
stweek4yash2026	Storage account	OK	Operation details

Next steps

Go to resource

Home > stweek4yash2026_1780314180664 | Overview

stweek4yash2026

Storage account

Improve storage account data protection Audit disaster recovery readiness for this storage account +1

Search

Upload Open in Explorer Delete Move Refresh Open in mobile CLI / PS Feedback

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Security + networking

Data management

Settings

Monitoring

Add or remove favorites by pressing Ctrl+Shift+F

Essentials

Resource group (move) rg-week4-adf

Location southeastasia

Subscription (move) Azure for Students

Subscription ID 5e564759-acce-49a8-b2db-0348ba0f8c5d

Disk state Available

Tags (edit) Add tags

Performance Standard

Replication Locally redundant storage (LRS)

Account kind StorageV2 (general purpose v2)

Provisioning state Succeeded

Created 6/1/2026, 5:14:22 PM

Properties Monitoring Capabilities (7) Recommendations (0) Tutorials Tools + SDKs

Blob service

Hierarchical namespace Disabled

Default access tier Hot

Security

Require secure transfer for REST API operations Enabled

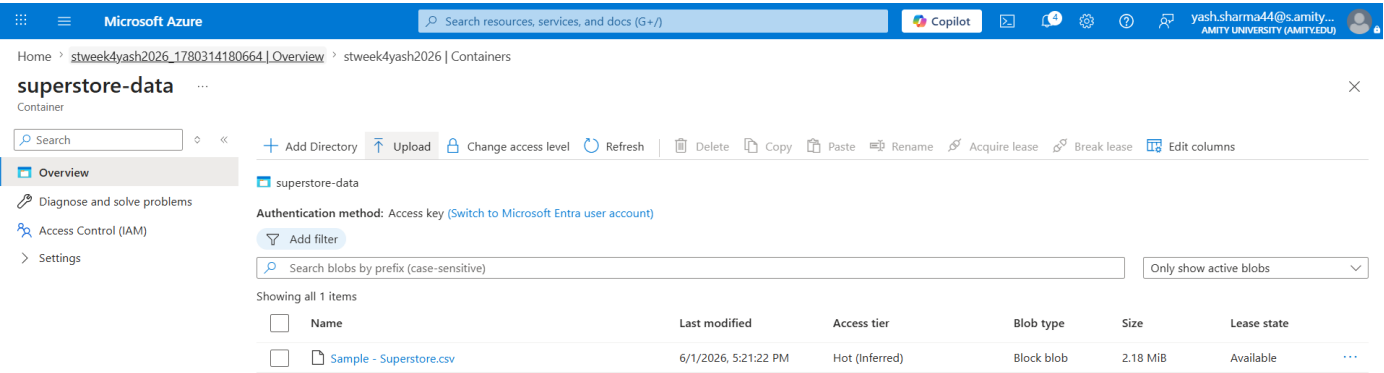
Storage account key access Enabled

Description:

I Created a Storage Account named stweek4yash2026 in the Southeast Asia region using the Azure for Students subscription. The Storage Account will be used to store files required for building the Azure Data Factory pipeline.

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Final Deliverable:



Description:

I created a Blob container named superstore-data inside the storage account and where finally I uploaded the Sample – Superstore.csv , as visible in the final screenshot. This file will be used as the source data for the Azure Data Factory pipeline.

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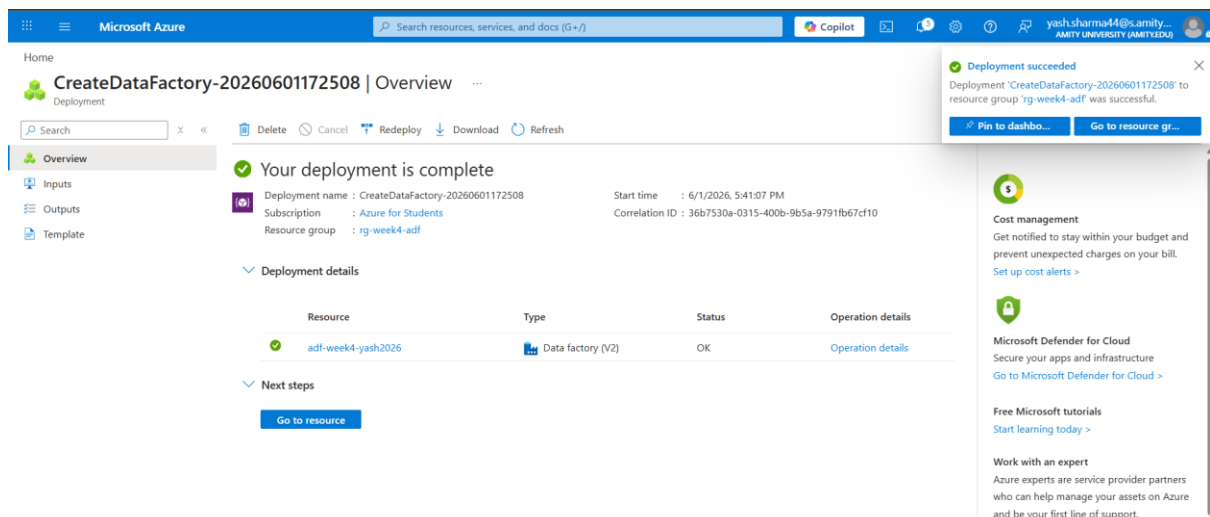
Task 3: ADF Basics

Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

Deliverable:

- Screenshot of Linked Service
- Screenshot of dataset
- Screenshot of Get Metadata activity

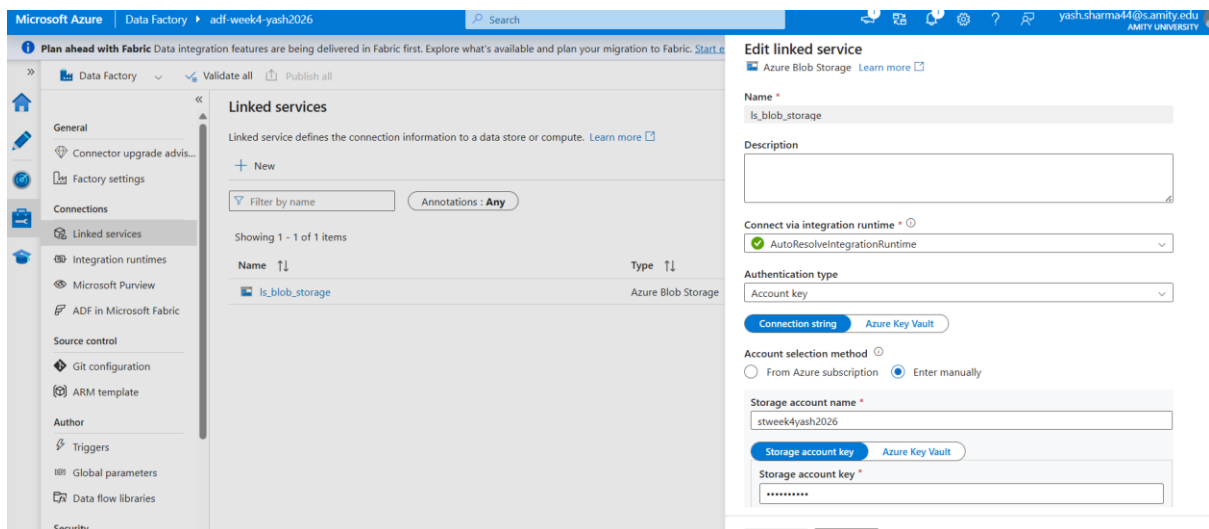


Description:

Created an Azure Data Factory instance named adf-week4-yash2026. As per the given portal assignment I have explored the main sections of the ADF Studio interface including Author, Monitor, and Manage. These sections are used for building, monitoring, and configuring data pipelines.

The screenshot displays the Microsoft Fabric Data Factory interface. At the top, a navigation bar includes the user's name 'yash.sharma44@samity.edu' and the 'AMITY UNIVERSITY' logo. Below the navigation bar, a banner message states: 'Integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)'. The main workspace is divided into several sections. On the left, the 'Data Factory' tab is active, showing a list of datasets: 'ds_superstore_csv', 'pl_superstore_pipe...', and 'ds_superstore_output'. The 'pl_superstore_pipe...' pipeline is selected, and its activities are displayed in the center. The activities are 'Get Metadata1' and 'Copy data'. The 'Copy data' activity is selected, and its properties are shown in the bottom pane. The properties include: Name 'Copy_Superstore_Data', Description, Activity state 'Activated', and Timeout '0.12:00:00'. On the right, the 'Properties' section is visible, showing the pipeline's Name 'pl_superstore_pipeline' and Description. The 'Annotations' section is also visible, with a '+ New' button.

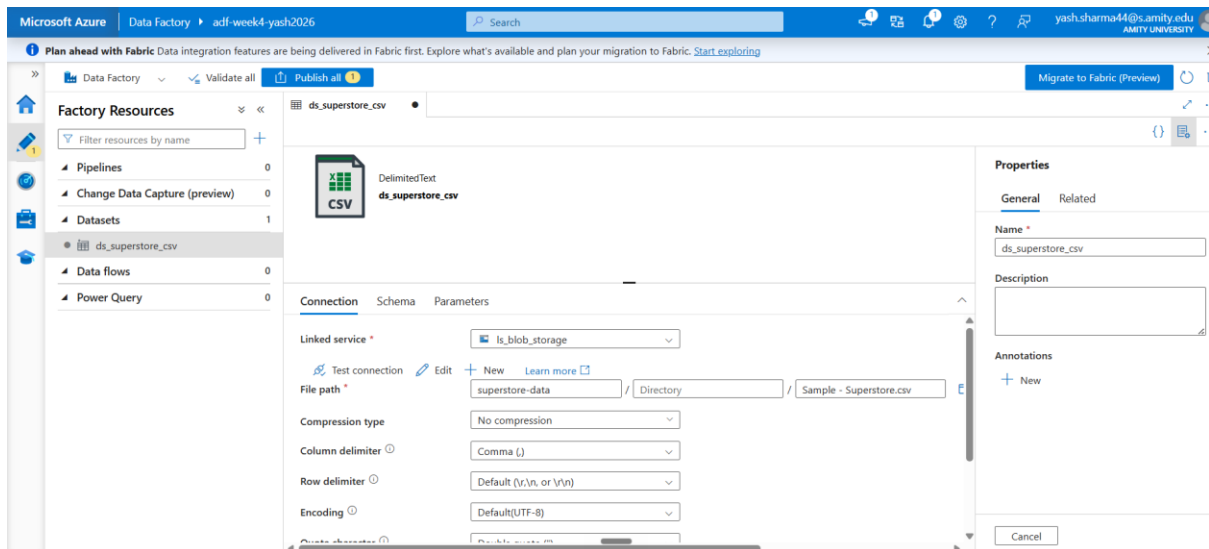
- Screenshot of Linked Service



I Created a Linked Service named ls_blob_storage to establish a connection between Azure Data Factory and Azure Blob Storage.

- Screenshot of dataset

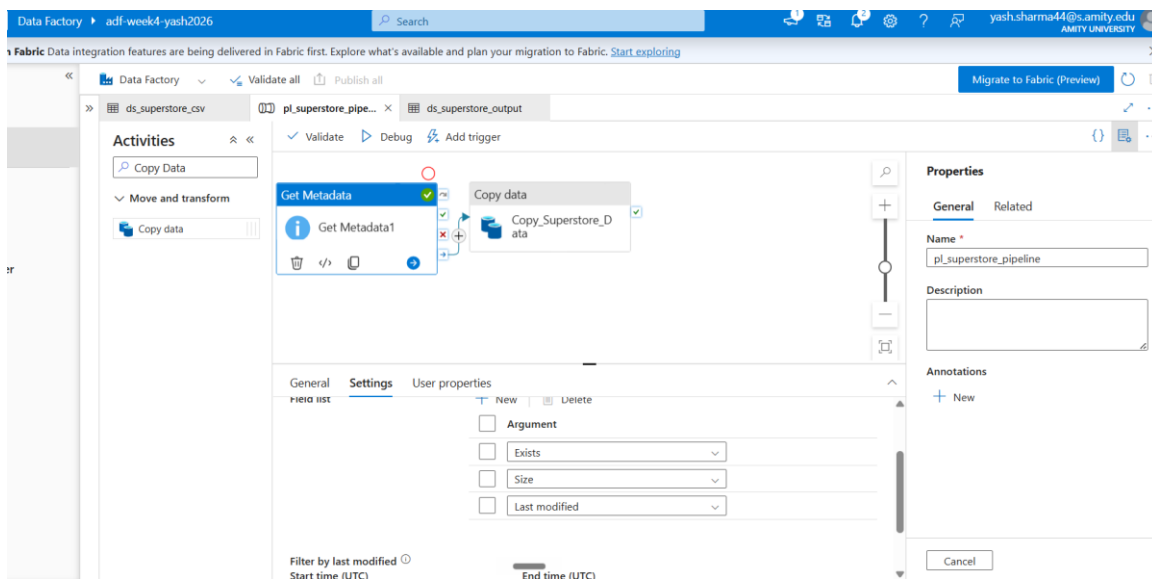
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Description:

I Created a dataset named ds_superstore_csv using the uploaded Superstore CSV file stored in Azure Blob Storage. The dataset will be used as the source for pipeline activities.

- Screenshot of Get Metadata activity



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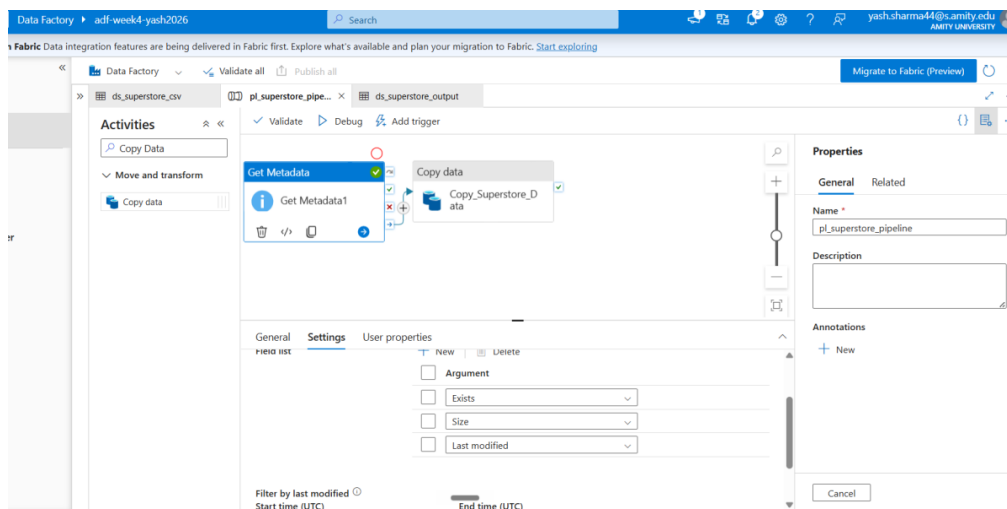
Task 4: Pipeline Development

Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

Deliverable:

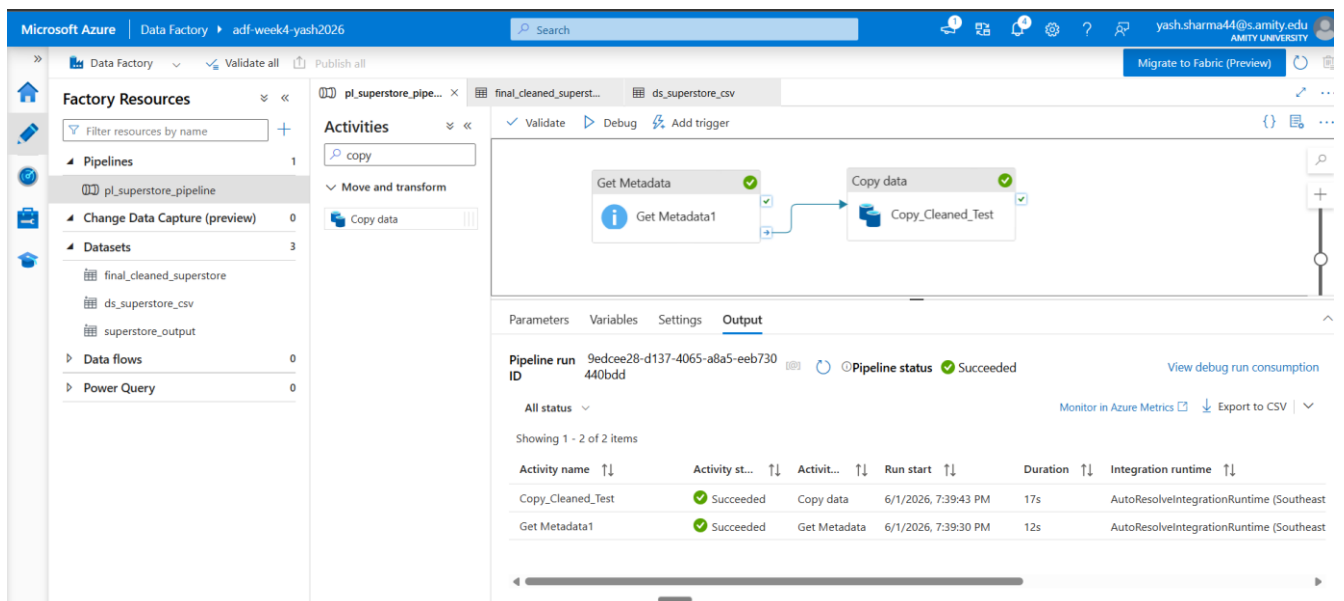
- Screenshot of pipeline design



Note : ForEach activity was not used because the pipeline till here, is processing a single CSV file.

Final Deliverable:

- Screenshot of pipeline design



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Task 5: Pipeline Execution

Do:

- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)

The screenshot displays the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane shows a list of pipelines, including 'pl_superstore_pipeline'. The main workspace shows the 'Copy data' activity, which is part of a pipeline named 'pl_superstore_pipeline'. The activity is connected to 'Get Metadata1' and 'Copy_Cleaned_Test'. The 'Output' tab at the bottom shows the pipeline run ID '9edcee28-d137-4065-a8a5-eeb730-440bdd' and the status 'Succeeded'. Below this, a table lists the activities and their execution details.

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime
Copy_Cleaned_Test	Succeeded	Copy data	6/1/2026, 7:39:43 PM	17s	AutoResolveIntegrationRuntime (Southeast)
Get Metadata1	Succeeded	Get Metadata	6/1/2026, 7:39:30 PM	12s	AutoResolveIntegrationRuntime (Southeast)

The screenshot shows the Microsoft Azure Storage Explorer interface. The 'superstore-data' container is selected, and the 'Overview' tab is active. The container contains three blobs: 'Sample - S...', 'final_cleane...', and 'superstore...'. The table below lists the details of these blobs.

Name	Last modified	Access tier	Blob type	Size
Sample - S...	6/1/2026, 5:21:22 PM	Hot (Inferred)	Block blob	2.18 MiB
final_cleane...	6/1/2026, 7:21:08 PM	Hot (Inferred)	Block blob	2.6 MiB
superstore...	6/1/2026, 7:39:57 PM	Hot (Inferred)	Block blob	2.6 MiB

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Observation:

After creating activities during pipeline execution, the Copy Data activity initially failed with the error `DelimitedTextMoreColumnsThanDefined`. After going through the documentation link and other options like step by step debugging and verifying the original Kaggle final then cleaning it as I did in week 3, I found that the issue was an incorrect Escape Character setting which arisen due to commas being in mid of consecutive “ “ double quotes. The source CSV used escaped double quotes (“”) within the text fields while I configured dataset with default backslash escaping. Updating the escape character to double quote resolved the parsing issue and allowed the pipeline to execute successfully.

Task 6: IAM Roles

Do:

- Assign roles:
 - Reader
 - Contributor
- Provide access to ADF for Storage

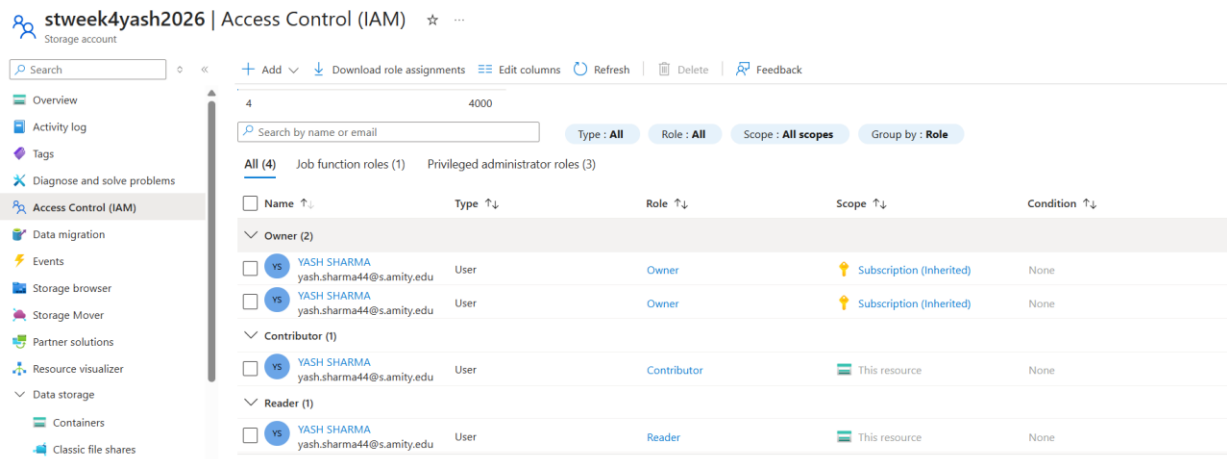
Deliverable:

- Screenshot of role assignment

The screenshot displays the 'Access Control (IAM)' page for a storage account named 'stweek4yash2026'. The 'Role assignments' tab is active, showing a list of roles assigned to the account. The page indicates 3 role assignments for this subscription. The roles are categorized into Job function roles (1) and Privileged administrator roles (2). The list shows two 'Owner' roles and one 'Reader' role, all assigned to the user 'YASH SHARMA' (yash.sharma44@s.amity.edu). The scope for the Owner roles is 'Subscription (Inherited)', and for the Reader role, it is 'This resource'.

Name	Type	Role	Scope	Condition
YASH SHARMA (yash.sharma44@s.amity.edu)	User	Owner	Subscription (Inherited)	None
YASH SHARMA (yash.sharma44@s.amity.edu)	User	Owner	Subscription (Inherited)	None
YASH SHARMA (yash.sharma44@s.amity.edu)	User	Reader	This resource	None

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Mini Project (Final Task)

Problem Statement:

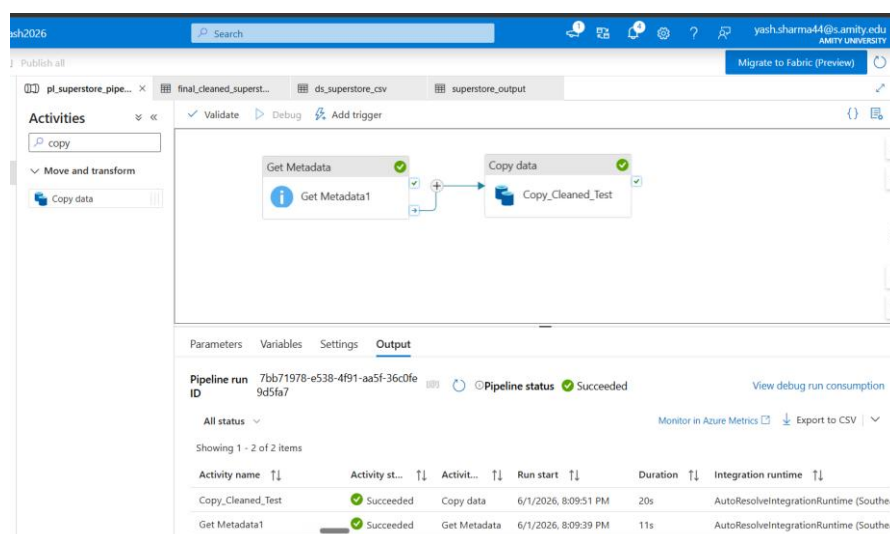
Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

- Pipeline executed successfully



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- Data copied to destination

Preview data

Linked service: ls_blob_storage
Object: superstore_output

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity
1	1	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.96	2
2	2	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	3
3	3	CA-2016-138688	06-12-2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Universal	14.62	2
																	Bretford		

- Metadata validated

Output

 Copy to clipboard

```
{
  "exists": true,
  "size": 2287806,
  "lastModified": "2026-06-01T11:51:22Z",
  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (Southeast Asia)",
  "executionDuration": 0,
  "durationInQueue": {
    "integrationRuntimeQueue": 4
  },
  "billingReference": {
    "activityType": "PipelineActivity",
    "billableDuration": [
      {
        "meterType": "AzureIR",
        "duration": 0.016666666666666666,
        "unit": "Hours"
      }
    ]
  }
}
```

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Output:

```
{
  "exists": true,
  "size": 2287806,
  "lastModified": "2026-06-01T11:51:22Z",
  "effectiveIntegrationRuntime": "AutoResolveIntegrationRuntime (Southeast Asia)",
  "executionDuration": 0,
  "durationInQueue": {
    "integrationRuntimeQueue": 4
  },
  "billingReference": {
    "activityType": "PipelineActivity",
    "billableDuration": [
      {
        "meterType": "AzureIR",
        "duration": 0.016666666666666666,
        "unit": "Hours"
      }
    ]
  }
}
```